

Meeting 1203

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December 03, 2025

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11.26 Discussion

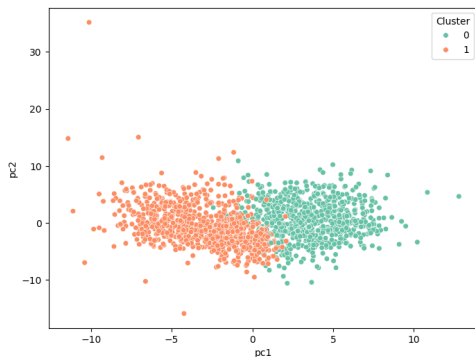
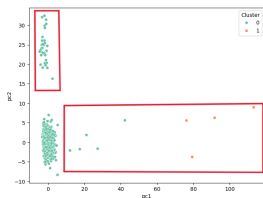
Semiconductor Manufacturing Process

- ▶ 抽掉 outlier 讓 PCA 更散更清楚
- ▶ (Y_i, X_i) iid $F(\cdot)$

11.26 Todo

- ▶ 繼續分成幾群再套模型比較最合適的
- ▶ 對 outlier 做質性的統計方法
- ▶ 在降維分群之前做 SMOTE 看看效果

Results I PCA + KMeans, 2 clusters



cluster	0	1
$y = 0$	660	770
$y = 1$	58	37

Results I

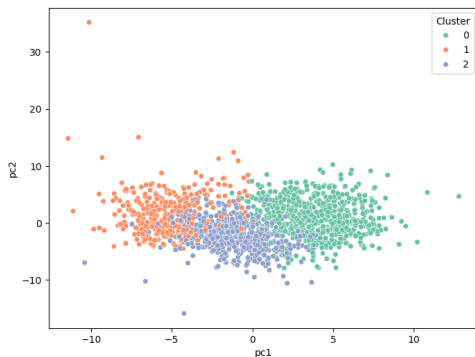
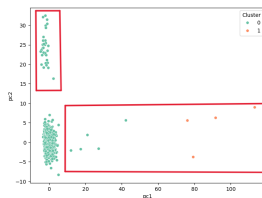
ns=10, 5-folds cv - mean

Summary

model	ACC	FAR	ROCAU
LR	0.8520	0.1011	0.5459
SVM	0.9304	0.0041	0.5027
DT	0.8149	0.1415	0.5309
RF	0.9304	0.0048	0.5074
XGB	0.9272	0.0096	0.5145

model	C ₀ ACC	C ₀ FAR	C ₀ ROCAU	C ₁ ACC	C ₁ FAR	C ₁ ROCAU
LR	0.8315	0.1152	0.5545	0.9009	0.0649	0.5586
SVM	0.9178	0.0015	0.4992	0.9542	0.0000	0.5000
DT	0.8273	0.1258	0.5848	0.8934	0.0753	0.5802
RF	0.9151	0.0045	0.4977	0.9554	0.0000	0.5143
XGB	0.9081	0.0197	0.5341	0.9554	0.0013	0.5261

Results II PCA + KMeans, 3 clusters



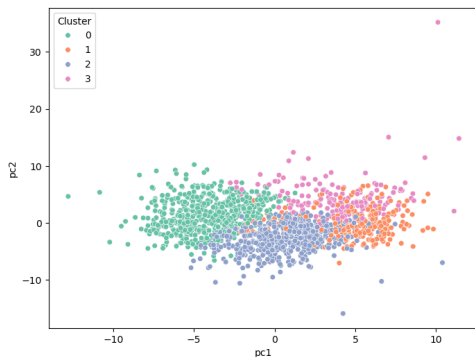
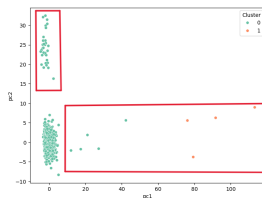
cluster	0	1	2
$y = 0$	592	327	511
$y = 1$	49	19	27

Results II ns=10, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8268	0.1149	0.5036	0.8930	0.0612	0.5278
SVM	0.9251	0.0000	0.5111	0.9422	0.0031	0.4985
DT	0.8315	0.1200	0.5633	0.8701	0.0856	0.5072
RF	0.9220	0.0017	0.4992	0.9451	0.0000	0.5000
XGB	0.9111	0.0169	0.5116	0.9335	0.0122	0.4939

model	C_2 ACC	C_2 FAR	C_2 ROCAU
LR	0.9183	0.0470	0.6032
SVM	0.9498	0.0000	0.5000
DT	0.8942	0.0705	0.5747
RF	0.9498	0.0000	0.5000
XGB	0.9386	0.0137	0.5098

Results III PCA + KMeans, 4 clusters



cluster	0	1	2	3
$y = 0$	513	231	523	163
$y = 1$	39	12	32	12

Results III

ns=3, 5-folds cv - mean

model	C_0 ACC	C_0 FAR	C_0 ROCAU	C_1 ACC	C_1 FAR	C_1 ROCAU
LR	0.8532	0.0937	0.5335	0.9095	0.0476	0.5095
SVM	0.9294	0.0000	0.5000	0.9507	0.0000	0.5000
DT	0.8224	0.1228	0.4886	0.8849	0.0821	0.5923
RF	0.9294	0.0000	0.5000	0.9507	0.0000	0.5000
XGB	0.9203	0.0136	0.5182	0.9466	0.0043	0.4978

model	C_2 ACC	C_2 FAR	C_2 ROCAU	C_3 ACC	C_3 FAR	C_3 ROCAU
LR	0.8973	0.0631	0.5947	0.8857	0.0803	0.6765
SVM	0.9423	0.0000	0.5000	0.9314	0.0000	0.5000
DT	0.8468	0.1069	0.4918	0.8514	0.0860	0.4570
RF	0.9423	0.0000	0.5000	0.9314	0.0000	0.5000
XGB	0.9423	0.0019	0.5133	0.9200	0.0123	0.4938

The End

Questions & Comments